Joint Before Centering at a Poisson-Long Base: Full-Rectangle Exact Conductors and High-β Initial-Value Closure

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Abstract

Let $P_{m,S}$ be the calibrated divisor prefix in a primitive opened row $m = \ell d$, and suppose that the base cutoff S may already exceed the orbit length $J \asymp X/Q$. The centered nonzero-spectrum theorem used previously requires $S < J$. Treating its scalar correction terms separately also incurs the polynomial loss $1 + S/J$. We remove both appearances of that restriction by changing the order of the decomposition.

Write

$$\mathsf{A}_{m,S}(j) = \sum_{\substack{u \le S \\ u \mid mj+h}} b_R(u), \qquad P_{m,S} = \mathsf{A}_{m,S} - \delta_H(m).$$

At additive frequency zero, the raw joint kernel, both drift legs, and the constant term recombine exactly as

$$\mathcal{R}^0_S(m,n) - \delta_H(m)D_{n,S} - \delta_H(n)D_{m,S} + \delta_H(m)\delta_H(n) = K^0_S(m,n) + E_S(m)E_S(n).$$

Here $D_{m,S} = \delta_H(m) + E_S(m)$, while K^0_S is the Ramanujan-square connected kernel already known to have arbitrary logarithmic physical saving. Thus no Poisson-short hypothesis is needed at zero frequency.

Away from zero, we prove a full-rectangle version of the symmetric exact-conductor argument. For $u, v \le S$, denominator $[u,v] \asymp Y$, and coordinate conductors F_1, F_2, the raw joint cell satisfies

$$\mathcal{L}^{\mathrm{base}}_{Y;S}(F_1,F_2) \ll QX^{\varepsilon}\sqrt{\mathcal{K}_Q(F_1)\mathcal{K}_Q(F_2)}\min\{\sqrt{JY}, F_1F_2\}, \qquad \mathcal{K}_Q(F) = F + Q/F.$$

The corresponding relaxed-box saving is the same piecewise function $M_\beta(s)$ that occurs in the annular joint problem. A pointwise target-divisor bound gives $\mathsf{A}_{m,S}(j) \ll_\varepsilon X^\varepsilon$, so the two remaining drift legs are power-small from the existing row-prime condition $L \ge X^{\lambda_0}$; no factor $1 + S/J$ remains. Consequently $M_\beta(s) > 0$, together with the physical row, no-wrap, Fourier-tail, and fixed-margin interfaces, closes the selected calibrated base packet even when $S > J$.

For the exact source-compatible sample

$$\beta = \frac{267}{400}, \qquad s = \frac{23}{60}, \qquad t = \frac{193}{500},$$

the old centered ledger has the negative face $-1/800$, whereas the full-rectangle base margin is $M_\beta(s) = 13/1600$. The row-prime exponent is $99979/210000$, and the inherited annular margin to $T = X^t$ is $33/8000$. Hence the selected calibrated truncated square closes at this particular $\beta > 2/3$ sample. The ultra-long complement, positivity, a Hardy–Littlewood asymptotic, twin primes, and any general breach of the parity barrier remain outside the result.

Keywords: primitive Möbius tail; Poisson-long base; raw joint rectangle; exact conductor; additive zero mode; sieve parity.

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1 Introduction

This paper addresses one precise initial-value gap in the primitive Möbius-tail program of TPC-18–27. The target is not a prime-pair asymptotic. It is a cancellation estimate for one selected calibrated quadratic packet that occurs before the ultra-long complement and before any positivity argument.

The previous chain separates into two facts. First, TPC-26 proves cutoff stability

$$\mathcal{Q}_T^{\mathrm{phys}} - \mathcal{Q}_S^{\mathrm{phys}} \ll_B XQ(\log X)^{-B} \tag{1}$$

in a sharp joint-conductor region that can contain $S > J$ and $T > J$ [6]. Second, TPC-27 proves the matching initial value

$$\mathcal{Q}_S^{\mathrm{phys}} \ll_B XQ(\log X)^{-B} \tag{2}$$

only when $S < J$ with a fixed power margin [7]. The logical difference matters: (1) alone does not estimate either endpoint.

The high-β sample recorded in TPC-26 has

$$\beta = \frac{267}{400}, \qquad J = X^{133/400+o(1)}, \qquad S = X^{23/60+o(1)}, \qquad T = X^{193/500+o(1)}. \tag{3}$$

Both cutoffs exceed J. Its annular margin is positive, but its initial value was therefore left open.

1.1 Why the earlier base proof stops

For a row $m = \ell d$, the calibrated prefix admits two exact representations:

$$P_{m,S} = Z_{m,S} + E_S(m) = \mathsf{A}_{m,S} - \delta_H(m). \tag{4}$$

The first uses a centered divisor process and a calibration error $E_S(m) \ll_A (\log X)^{-A}$. The second uses the raw divisor prefix $\mathsf{A}_{m,S}$ and the row drift $\delta_H(m) \ll X^\varepsilon/L$.

TPC-27 uses the centered representation at every frequency. Its nonzero exact-conductor ledger is

$$\eta_S(\beta, s) = \min\left\{1 - \frac{3\beta}{2}, \frac{\beta + 1 - 4s}{2}, \frac{3 - \beta - 6s}{4}\right\}. \tag{5}$$

At (3), its first entry is $-1/800$. Moreover, estimating $E_S Z_S$ by an orbitwise ℓ^1 bound introduces $J + S$. An error that is smaller than every logarithmic power cannot absorb the fixed power S/J.

Neither obstruction occurs if the two representations in (4) are used at different additive frequencies. This is the central observation of the paper.

1.2 The frequency-dependent recombination

At additive frequency zero we keep the centered representation. The complete-period mean of $Z_{m,S}$ vanishes, so the zero kernel of $P_{m,S}P_{n,S}$ is exactly

$$K_S^0(m, n) + E_S(m)E_S(n). \tag{6}$$

The potentially dangerous EZ and ZE terms are absent by exact algebra, not by an estimate. TPC-27 already closes K_S^0 for every polynomial $S \geq R$; its zero theorem never required $S < J$.

At nonzero frequency we instead expand

$$P_{m,S}P_{n,S} = \mathsf{A}_{m,S}\mathsf{A}_{n,S} - \delta_H(m)\mathsf{A}_{n,S} - \delta_H(n)\mathsf{A}_{m,S} + \delta_H(m)\delta_H(n). \tag{7}$$

The coefficient-one term is one raw joint rectangle $u, v \le S$. The other three terms contain the fixed-power row drift. On the physical target support, a divisor bound gives

$$\mathsf{A}_{m,S}(j) \ll_\varepsilon X^\varepsilon, \tag{8}$$

uniformly in S. Hence the full drift contribution is already power-small from $L \ge X^{\lambda_0}$; no $J+S$ loss is needed.

For the raw joint rectangle we prove the full-base analogue of the TPC-26 annular cell theorem. The annular lower restriction was deleted when enlarging the nonnegative tensor energy, but the full-base result is not obtained merely by setting an endpoint to zero. We define the full fibers, their exact-conductor projections, principal mean, and Fourier reassembly anew. The resulting cell estimate is

$$\mathcal{L}^{\mathrm{base}}_{Y;S}(F_1, F_2) \ll_\varepsilon QX^\varepsilon \sqrt{\mathcal{K}_Q(F_1)\mathcal{K}_Q(F_2)}\, \min\{\sqrt{JY}, F_1F_2\}. \tag{9}$$

It gives the sharp relaxed-box margin $M_\beta(s)$, without the old face $1 - 3\beta/2$.

1.3 Main consequence

Under the actual opened-row provenance, the divisor-independent off-diagonal mask, the common physical weight, no-wrap, and fixed Fourier-tail interfaces, we prove

$$\boxed{M_\beta(s) > 0 \quad \Longrightarrow \quad \mathcal{Q}^{\mathrm{phys}}_S \ll_B XQ(\log X)^{-B}} \tag{10}$$

for every fixed $B > 0$. There is no condition $S < J$. At the high sample,

$$M_{267/400}(23/60) = \frac{13}{1600} > 0. \tag{11}$$

Combining (10) with the TPC-26 annular margin

$$M_{267/400}(193/500) = \frac{33}{8000} > 0 \tag{12}$$

closes the selected calibrated truncated square through T.

The improvement is a recombination theorem inside a specific analytic packet. It does not estimate the complement beyond T, whose divisor reflection exposes polynomial families of affine two-point Möbius correlations [4, 6]. It also does not produce a positive main term. In particular, the result neither implies twin primes nor claims a general breach of the sieve parity barrier.

1.4 Organization

Section 2 fixes the physical interface and proves the exact zero-frequency recombination. Section 3 constructs the full-rectangle exact-conductor cells and proves the two cell estimates. Section 4 optimizes and reassembles the nonzero spectrum. Section 5 restores the drift and proves the Poisson-long base theorem. Section 6 checks the exact high-β ledger and splices to TPC-26. Section 7 records the certificate, dependency ledger, and scope.

2 Physical interface and exact zero recombination

We retain the physical packet before Mellin separation. This section contains the only zero-frequency argument in the paper.

2.1 Rows, scales, and coefficients

Fix $h \in \mathbb{Z} \setminus \{0\}$, and put

$$H = \operatorname{rad}|h|, \qquad H_0 = |h|. \tag{13}$$

Let

$$R = X^{r_0+o(1)}, \qquad r_0 = \frac{1}{2} - \delta > 0, \qquad R \le S \le X^{C_S}. \tag{14}$$

Rows in one opened slice are

$$\alpha = (\ell, d_\alpha), \qquad m_\alpha = \ell d_\alpha \asymp Q = LD, \tag{15}$$

where $\ell \asymp L$ is prime, $d_\alpha \asymp D$ is squarefree, $(d_\alpha, H) = 1$, and

$$d_\alpha \le R^{1/2}, \qquad QJ \asymp X, \qquad L > \max\{2R, 2H_0\}, \qquad L \ge X^{\lambda_0} \tag{16}$$

for a fixed $\lambda_0 > 0$. All exponent inequalities have fixed positive margins. The physical orbit support is chosen so that

$$|m_\alpha j + h| \asymp X \tag{17}$$

whenever its smooth weight is nonzero. In particular the target is never zero.

On the primitive orbit $(j, H) = 1$, divisibility gives

$$(u, m_\alpha H) = 1 \quad \text{whenever} \quad u \mid m_\alpha j + h. \tag{18}$$

The two physical row coefficients have the opened-d provenance

$$\gamma_\alpha^{(i)} = \mu(d_\alpha)(\log \ell)\omega_D^{(i)}(d_\alpha)\psi_L^{(i)}(\ell/L)\zeta_\alpha^{(i)}, \qquad |\zeta_\alpha^{(i)}| \ll 1, \tag{19}$$

and satisfy

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll Q, \tag{20}$$

$$\|\gamma^{(i)}\|_2^2 \ll Q \log(2L). \tag{21}$$

These are the physical provenance bounds isolated in TPC-25 [5].

The mask $\mathfrak{m}(\alpha, \gamma)$ is bounded by one, independent of every divisor and conductor variable, and vanishes when $m_\alpha = m_\gamma$. The periodic factor $\Xi_{\alpha,\gamma}(j)$ has fixed period H_0. The smooth row–orbit weight $\mathcal{W}_{\alpha,\gamma}(j/J)$ belongs to the same factorable bounded C_c^∞ family as in TPC-25–27. Its support has $O(J)$ integer points, its zero Fourier functional is uniformly bounded, and its Mellin representation has finite weighted variation of every fixed order. All fixed source, residue, smoothness, and margin data are denoted by $\mathscr{D}$.

Put

$$a(u) = -\mu(u)\log u, \qquad \lambda_R'(u) = \begin{cases} u \displaystyle\sum_{b \le R/u} \frac{\mu(ub)\mu(b)}{\varphi(ub)}, & u \le R, \\ 0, & u > R, \end{cases} \qquad b_R(u) = a(u) - \lambda_R'(u). \tag{22}$$

All three sequences are supported on squarefree integers. The elementary positive majorant used in TPC-24 and TPC-27 gives

$$|b_R(u)| \ll_{\mathscr{D}} (\log(2X))^3 \qquad (u \le X^{C_S}). \tag{23}$$

Indeed, $u/\varphi(u) \ll \log\log(3X)$ and $\sum_{b \le R/u} \mu^2(b)/\varphi(b) \ll \log(2X)$; the displayed third power is a convenient uniform envelope.

For $m = \ell d$, define the row drift

$$\delta_H(m) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)(\ell - 1)} \ll_{\varepsilon,\mathscr{D}} \frac{X^\varepsilon}{L}. \tag{24}$$

2.2 Raw, centered, and calibrated prefixes

For $R \le S$, let

$$\mathsf{A}_{m,S}(j) := \sum_{\substack{u\le S\\ u\mid mj+h}} b_R(u), \tag{25}$$

$$D_{m,S} := \sum_{\substack{u\le S\\ (u,mH)=1}} \frac{b_R(u)}{u}, \tag{26}$$

$$Z_{m,S}(j) := \sum_{\substack{u\le S\\ (u,mH)=1}} b_R(u)\left(\mathbf{1}_{u\mid mj+h} - \frac{1}{u}\right), \tag{27}$$

$$P_{m,S}(j) := \mathsf{A}_{m,S}(j) - \delta_H(m). \tag{28}$$

The finite-model density identity and the proved uniform Möbius calibration theorem give the exact relation

$$D_{m,S} = \delta_H(m) + E_S(m), \qquad E_S(m) \ll_{A,\mathscr{D}} (\log X)^{-A} \tag{29}$$

for every fixed $A > 0$, uniformly in the retained polynomial row and cutoff ranges [2, 7]. Consequently

$$\boxed{P_{m,S} = \mathsf{A}_{m,S} - \delta_H(m) = Z_{m,S} + E_S(m).} \tag{30}$$

The physical base packet is

$$\begin{aligned}\mathcal{Q}_S^{\mathrm{phys}} := &\sum_{\alpha,\gamma} \gamma_\alpha^{(1)}\gamma_\gamma^{(2)}\mathfrak{m}(\alpha,\gamma)\\ &\times \sum_{\substack{j\in\mathbb{Z}\\ (j,H)=1}} \Xi_{\alpha,\gamma}(j)\mathcal{W}_{\alpha,\gamma}(j/J)P_{m_\alpha,S}(j)P_{m_\gamma,S}(j).\end{aligned} \tag{31}$$

Its natural scale is

$$JQ^2 \asymp XQ. \tag{32}$$

2.3 The exact zero-frequency identity

For distinct rows m, n, put

$$\mathcal{R}_S^0(m,n) := \sum_{\substack{u,v\le S\\ (u,mH)=1\\ (v,nH)=1}} \frac{b_R(u)b_R(v)}{uv}(u,v)\mathbf{1}_{(u,v)\mid m-n}, \tag{33}$$

$$K_S^0(m,n) := \sum_{\substack{u,v\le S\\ (u,mH)=1\\ (v,nH)=1}} \frac{b_R(u)b_R(v)}{uv}\left((u,v)\mathbf{1}_{(u,v)\mid m-n} - 1\right). \tag{34}$$

The fixed outer primitive density is suppressed in both definitions; it is restored once in the physical Poisson coefficient. The same formulas hold inside a fixed residue class modulo H_0, up to one bounded local factor.

Proposition 2.1 (Raw-to-centered zero recombination)**.** *For every retained pair of rows,*

$$\mathcal{R}_S^0(m,n) = K_S^0(m,n) + D_{m,S}D_{n,S}, \tag{35}$$

$$\mathcal{R}_S^0(m,n) - \delta_H(m)D_{n,S} - \delta_H(n)D_{m,S} + \delta_H(m)\delta_H(n) = K_S^0(m,n) + E_S(m)E_S(n). \tag{36}$$

After factoring out the fixed outer primitive density, the left side of (36) is exactly the complete-period arithmetic zero kernel of $P_{m,S}P_{n,S}$.

Proof. The primitive CRT density of the two target congruences is

$$\frac{(u,v)}{uv}\mathbf{1}_{(u,v)|m-n}, \tag{37}$$

after suppressing the fixed outer density. This proves that (33) is the raw zero kernel. Subtracting (34) leaves

$$\sum_u \frac{b_R(u)}{u}\sum_v \frac{b_R(v)}{v} = D_{m,S}D_{n,S},$$

with the row-coprimality restrictions in the corresponding factors. This proves (35). Insert $D_{m,S} = \delta_H(m) + E_S(m)$ and expand. The two mixed terms and the $\delta_H(m)\delta_H(n)$ term cancel exactly, giving (36). The last assertion follows by taking the complete-period mean in $P_{m,S} = \mathsf{A}_{m,S} - \delta_H(m)$. □

Remark 2.2 (Why the order matters). If one first takes absolute values of the three scalar channels in $(Z+E)(Z+E)$, the exact cancellation in (36) is lost. The identity is a frequency-zero statement for the complete calibrated product, not an estimate for its four expanded pieces.

2.4 Physical zero closure

TPC-27 proves that for every fixed $A > 0$, every fixed

$$0 < \kappa < \min\{s, r_0/2\}, \qquad S = X^{s+o(1)}, \tag{38}$$

and distinct retained rows,

$$K_S^0(m,n) \ll_{A,\mathscr{D}} (\log X)^{-A} + L^{-2}(\log\log(3X))^3 + X^{-\kappa+o(1)}. \tag{39}$$

This is the Ramanujan-square theorem; it has no hypothesis $S < J$ [7].

Theorem 2.3 (Physical calibrated zero closure)**.** *Retain the physical interface above, the parameter range* (38)*, and the imported bound* (39)*. Then the additive zero contribution of the complete product in* (31) *satisfies, for every fixed* $B > 0$*,*

$$\boxed{(\mathcal{Q}_S^{\mathrm{phys}})^0 \ll_{B,\mathscr{D}} XQ(\log X)^{-B}.} \tag{40}$$

No Poisson-short condition is assumed.

Proof. Split the fixed H_0-periodic factor into its finitely many residue classes and take physical frequency zero before Mellin separation. The zero functional of the smooth weight and the fixed local density are uniformly bounded. By proposition 2.1, the arithmetic kernel is $K_S^0(m,n) + E_S(m)E_S(n)$. The mask deletes $m = n$, so (39) applies. Delete the bounded mask and zero functional, then use (20) twice. The result is

$$|(\mathcal{Q}_S^{\mathrm{phys}})^0| \ll JQ^2\{(\log X)^{-A} + L^{-2}(\log\log(3X))^3 + X^{-\kappa+o(1)} + (\log X)^{-2A}\}.$$

Now use (32), choose A after B, and use $L \geq X^{\lambda_0}$ and $\kappa > 0$. This proves the theorem. □

3 The full-base exact-conductor rectangle

We now treat the nonzero additive spectrum of the coefficient-one term $\mathsf{A}_{m,S}\mathsf{A}_{n,S}$ in (7). All divisor pairs $u, v \leq S$ are retained. There is no annular lower cutoff.

3.1 Physical separation after zero extraction

After the zero term of theorem 2.3 has been removed, split the fixed H_0-periodic multiplier into its finitely many residue cells. On one cell the physical coefficient–weight product has the continuous representation

$$\gamma_\alpha^{(1)}\gamma_\gamma^{(2)}\mathcal{W}_{\alpha,\gamma}(j/J) = \int x_{\boldsymbol{\tau}}(\alpha)y_{\boldsymbol{\tau}}(\gamma)\Phi_{\boldsymbol{\tau}}(j/J)\,d\nu(\boldsymbol{\tau}), \tag{41}$$

where every fixed weighted moment satisfies

$$\int (1+|\boldsymbol{\tau}|)^M \|x_{\boldsymbol{\tau}}\|_2\|y_{\boldsymbol{\tau}}\|_2\,d|\nu| \ll_{M,\mathscr{D}} Q\log(2L), \tag{42}$$

$$\int (1+|\boldsymbol{\tau}|)^M \|x_{\boldsymbol{\tau}}\|_1\|y_{\boldsymbol{\tau}}\|_1\,d|\nu| \ll_{M,\mathscr{D}} Q^2. \tag{43}$$

The row scales, target primitivity, and divisor-independent mask are unchanged. This is the physical transfer lemma of TPC-25–27 applied to the same provenance. We prove the conductor estimates for one component and integrate only after reassembly.

Fix one parameter $\boldsymbol{\tau}$ and abbreviate

$$x = x_{\boldsymbol{\tau}}, \qquad y = y_{\boldsymbol{\tau}}, \qquad G = \Phi_{\boldsymbol{\tau}}. \tag{44}$$

The Schwartz seminorms of G grow at most polynomially in $1+|\boldsymbol{\tau}|$, and those powers are absorbed by the weighted moments in equations (42) and (43).

Write

$$e(t) = e^{2\pi i t}, \qquad e_q(t) = e(t/q), \qquad \widehat{G}(y) = \int_{\mathbb{R}} G(x)e(-xy)\,dx. \tag{45}$$

For an effective squarefree pair, put

$$g = (u,v), \qquad u = gc, \qquad v = gd, \qquad q = [u,v] = g\,c\,d. \tag{46}$$

Then g, c, d are pairwise coprime. The target congruences are compatible exactly when

$$g \mid m-n. \tag{47}$$

When compatible they determine one unit class $\rho_{m,n}^{u,v} \pmod q$. Suppressing the harmless fixed residue cell, Poisson summation gives

$$\begin{aligned}\sum_{\substack{j\in\mathbb{Z}\\(j,H)=1}} &G(j/J)\mathbf{1}_{u|mj+h}\mathbf{1}_{v|nj+h}\\ &= \frac{J}{Hq}\sum_{r\in\mathbb{Z}} c_H(r)e_q(r\overline{H}\rho_{m,n}^{u,v})\widehat{G}\left(\frac{rJ}{Hq}\right),\end{aligned} \tag{48}$$

where $\overline{H}$ is the inverse of $H \pmod q$. The term $r=0$ has already been treated.

3.2 Pair fibers and exact coordinate conductors

From this point onward, row labels are written as m, n; the symbols α, γ denote multiplicative characters. Let $\mathfrak{X}(w)$ denote the character group of $(\mathbb{Z}/w\mathbb{Z})^\times$. Since $q = ud$ and $(u,d)=1$,

$$\mathfrak{X}(q) \simeq \mathfrak{X}(u)\times\mathfrak{X}(d), \qquad \chi = \alpha\otimes\gamma, \qquad \operatorname{cond}\chi = (\operatorname{cond}\alpha)(\operatorname{cond}\gamma). \tag{49}$$

For separated row weights x_m, y_n, define

$$c_{m,n}^{(g)} = x_m y_n \mathfrak{m}(m,n)\mathbf{1}_{g|m-n}. \tag{50}$$

The unit fiber and its discrepancy are

$$Z_{u,v}(\xi) := \sum_{\substack{m,n \\ g|m-n \\ \rho^{u,v}_{m,n}\equiv\xi \bmod q}} x_m y_n \mathfrak{m}(m,n), \tag{51}$$

$$\overline{Z}_{u,v} := \frac{1}{\varphi(q)} \sum_{\xi\in(\mathbb{Z}/q\mathbb{Z})^\times} Z_{u,v}(\xi), \qquad D_{u,v} = Z_{u,v} - \overline{Z}_{u,v}. \tag{52}$$

Finite CRT algebra gives the exact multiplicative transform

$$\widehat{Z}^{\times}_{u,v}(\alpha\otimes\gamma) = \overline{(\alpha\otimes\gamma)(-h)} \sum_{m,n} c^{(g)}_{m,n}\alpha(m)\gamma(n). \tag{53}$$

For every nonprincipal pair, centering changes no transform.

Let $\mathfrak{X}_F(w)$ be the characters modulo w whose exact conductor lies in a dyadic interval $\asymp F$, with $F = 1$ denoting the principal character. The pairwise conductor projection is

$$D^{F_1,F_2}_{u,v}(\xi) := \frac{1}{\varphi(q)} \sum_{\alpha\in\mathfrak{X}_{F_1}(u)} \sum_{\gamma\in\mathfrak{X}_{F_2}(d)} \widehat{D}^{\times}_{u,v}(\alpha\otimes\gamma)(\alpha\otimes\gamma)(\xi). \tag{54}$$

The off-diagonal mask gives the compatibility-energy bound

$$\sum_{g\le S}\sum_{m,n} |c^{(g)}_{m,n}|^2 \ll_\varepsilon X^\varepsilon \|x\|_2^2\|y\|_2^2. \tag{55}$$

Indeed, after reversing the order, the multiplicity is bounded by $\tau(|m-n|) \ll_\varepsilon X^\varepsilon$. This is the only use of $m \neq n$ in the nonzero energy.

Put

$$\mathcal{K}_Q(F) = F + \frac{Q}{F}. \tag{56}$$

For dyadic Y, F_1, F_2, define the full-base tensor energy

$$\begin{aligned}\mathcal{E}^{\mathrm{base}}_{Y;S}(F_1,F_2) := &\sum_{\substack{g,c,d \text{ pairwise coprime} \\ gc,gd\le S \\ g\,c\,d\asymp Y}} \frac{|b_R(gc)b_R(gd)|^2}{\varphi(g\,c\,d)} \\ &\times \sum_{\alpha\in\mathfrak{X}_{F_1}(gc)} \sum_{\gamma\in\mathfrak{X}_{F_2}(d)} \left|\sum_{m,n} c^{(g)}_{m,n}\alpha(m)\gamma(n)\right|^2. \end{aligned}\tag{57}$$

Theorem 3.1 (Full-base tensor conductor energy)**.** *Uniformly for $Y \le S^2$ and $F_1, F_2 \le S$,*

$$\boxed{\mathcal{E}^{\mathrm{base}}_{Y;S}(F_1,F_2) \ll_\varepsilon X^\varepsilon \|x\|_2^2\|y\|_2^2 \mathcal{K}_Q(F_1)\mathcal{K}_Q(F_2).} \tag{58}$$

Proof. Fix g and split both ambient moduli dyadically. Enlarge the nonnegative sum by deleting the divisibility and coprimality restrictions after the dyadic ranges have been retained. Apply the exact-conductor multiplicative large sieve in the following normalization, where (a_r) is supported in an interval of length N:

$$\sum_{\substack{w\asymp W \\ w \text{ squarefree}}} \frac{1}{\varphi(w)} \sum_{\substack{\chi \bmod w \\ \mathrm{cond}\,\chi\asymp F}} \left|\sum_r a_r\chi(r)\right|^2 \ll_\varepsilon X^\varepsilon \left(F + \frac{N}{F}\right) \sum_r |a_r|^2, \tag{59}$$

Here $F = 1$ denotes the principal character induced from conductor one; the case $w = 1$ is included and follows directly from Cauchy. Apply this inequality first in the m-coordinate in

Hilbert-valued form and then in the n-coordinate [1, 3, 6]. Both row intervals have length $O(Q)$, so the contribution for fixed g is

$$\ll_\varepsilon X^\varepsilon \mathcal{K}_Q(F_1)\mathcal{K}_Q(F_2) \sum_{m,n} |c_{m,n}^{(g)}|^2.$$

The pointwise bound (23), the logarithmically many dyadic subdivisions, and squarefree multiplicities are absorbed by X^ε. Sum over g and apply (55). □

Remark 3.2 (Why this is a full-base theorem). The energy proof resembles the annular proof because that proof also deleted its lower support restriction when enlarging a nonnegative sum. Nevertheless, (57) is a newly defined full rectangle. No illegal substitution of a cutoff below R is made in an annular prefix identity.

3.3 Two estimates for one nonzero cell

Assume the no-wrap margin

$$J \geq 4HX^{\varepsilon_0} \tag{60}$$

for a fixed $\varepsilon_0 > 0$, and choose $0 < \omega < \varepsilon_0$. For one divisor pair define the retained scalar

$$\mathcal{L}_{u,v}^{F_1,F_2,\leq\omega} := \frac{J}{Hq} \sum_{0<|r|\leq X^\omega Hq/J} c_H(r)\widehat{G}\left(\frac{rJ}{Hq}\right) \times \sum_{\xi\in(\mathbb{Z}/q\mathbb{Z})^\times} D_{u,v}^{F_1,F_2}(\xi)e_q(r\overline{H}\xi). \tag{61}$$

Aggregate it over the full rectangle by

$$\mathcal{L}_{Y;S,\boldsymbol{\tau}}^{\text{base}}(F_1,F_2) := \sum_{\substack{g,c,d \text{ pairwise coprime} \\ gc,gd\leq S \\ g\,c\,d\asymp Y}} |b_R(gc)b_R(gd)||\mathcal{L}_{gc,gd}^{F_1,F_2,\leq\omega}|. \tag{62}$$

For some fixed $C_{\mathscr{D}} > 0$, the polynomial seminorm growth described after (44) gives the following two homogeneous bounds.

Lemma 3.3 (Restricted-frequency norm route)**.** *For every exact-conductor cell,*

$$\mathcal{L}_{Y;S,\boldsymbol{\tau}}^{\text{base}}(F_1,F_2) \ll_{\varepsilon,\mathscr{D}} X^\varepsilon(1+|\boldsymbol{\tau}|)^{C_{\mathscr{D}}}\|x\|_2\|y\|_2\sqrt{JY\mathcal{K}_Q(F_1)\mathcal{K}_Q(F_2)}. \tag{63}$$

Proof. For one q, let

$$\mathcal{P}_q(\xi) = \frac{J}{Hq} \sum_{0<|r|\leq X^\omega Hq/J} c_H(r)\widehat{G}\left(\frac{rJ}{Hq}\right) e_q(r\overline{H}\xi).$$

The no-wrap condition makes the retained frequencies distinct modulo q. Additive orthogonality on all residues and Schwartz decay give

$$\sum_{\xi\in(\mathbb{Z}/q\mathbb{Z})^\times} |\mathcal{P}_q(\xi)|^2 \ll_{\mathscr{D}} (1+|\boldsymbol{\tau}|)^{C_{\mathscr{D}}} J. \tag{64}$$

Multiplicative Parseval then bounds one pair by

$$|\mathcal{L}_{u,v}^{F_1,F_2,\leq\omega}| \ll_{\mathscr{D}} (1+|\boldsymbol{\tau}|)^{C_{\mathscr{D}}} \left\{\frac{J}{\varphi(q)} \sum_{\alpha,\gamma} |\widehat{D}_{u,v}^\times(\alpha\otimes\gamma)|^2\right\}^{1/2}, \tag{65}$$

where the characters lie in the displayed conductor cells. There are $O_\varepsilon(YX^\varepsilon)$ pairwise-coprime triples with $g\,c\,d \asymp Y$. Apply Cauchy–Schwarz over the triples and then theorem 3.1. □

For the second route, let the exact conductors be f_1, f_2, put $f = f_1 f_2$, and write $q = ef$. Squarefreeness gives $(e, f) = 1$. The induced Gauss sum satisfies

$$\left| \sum_{\xi \in (\mathbb{Z}/q\mathbb{Z})^\times} (\alpha \otimes \gamma)(\xi) e_q(r\overline{H}\xi) \right| \le (f_1 f_2)^{1/2} |c_e(r)|. \tag{66}$$

At $r \neq 0$,

$$\sum_{r \neq 0} |c_H(r) c_e(r)| \left| \widehat{G}\left(\frac{rJ}{Hq}\right) \right| \ll_{\mathscr{D}} (1 + |\boldsymbol{\tau}|)^{C_{\mathscr{D}}} \frac{Hq}{J} \tau(e). \tag{67}$$

Lemma 3.4 (Induced-Gauss route)**.** *For every nonprincipal exact-conductor cell,*

$$\mathcal{L}^{\text{base}}_{Y;S,\boldsymbol{\tau}}(F_1, F_2) \ll_{\varepsilon,\mathscr{D}} X^{\varepsilon} (1 + |\boldsymbol{\tau}|)^{C_{\mathscr{D}}} \|x\|_2 \|y\|_2 F_1 F_2 \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \tag{68}$$

Proof. Insert (54) into (61). The elementary exact-conductor count is $\#\mathfrak{X}_{F_i}(w) \ll_\varepsilon F_i X^\varepsilon$ for squarefree w. Use (66) and (67), followed by Cauchy–Schwarz in the characters. Sum over divisor triples using Cauchy–Schwarz with weights $|b_R(u) b_R(v)|^2/\varphi(q)$ and $1/\varphi(q)$. The second weight has total $O_\varepsilon(X^\varepsilon)$ on $q \asymp Y$, since the number of pairwise-coprime factorizations is at most $3^{\omega(q)}$. The first is bounded by theorem 3.1. □

Theorem 3.5 (Full-rectangle cell bound)**.** *For every nonprincipal conductor cell,*

$$\mathcal{L}^{\text{base}}_{Y;S,\boldsymbol{\tau}}(F_1, F_2) \ll_{\varepsilon,\mathscr{D}} X^{\varepsilon} (1 + |\boldsymbol{\tau}|)^{C_{\mathscr{D}}} \|x_{\boldsymbol{\tau}}\|_2 \|y_{\boldsymbol{\tau}}\|_2 \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \min\{\sqrt{JY}, F_1 F_2\}. \tag{69}$$

Consequently,

$$\boxed{\int \mathcal{L}^{\text{base}}_{Y;S,\boldsymbol{\tau}}(F_1, F_2)\, d|\nu|(\boldsymbol{\tau}) \ll_{\varepsilon,\mathscr{D}} Q X^{\varepsilon} \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \min\{\sqrt{JY}, F_1 F_2\}.} \tag{70}$$

The global principal pair $(F_1, F_2) = (1, 1)$ is excluded; cells with exactly one principal coordinate are included.

Proof. Take the minimum of lemmas 3.3 and 3.4 to obtain (69). Integrate once and use (42); the factor $\log(2L)$ is absorbed by X^ε. This proves (70). □

4 Sharp base minimax and nonzero reassembly

We optimize theorem 3.5 over the complete base box and then restore the principal mean, Fourier tail, fixed residue cells, and physical Mellin components.

4.1 The exponent ledger

Write

$$Q = X^{\beta + o(1)}, \qquad J = X^{1 - \beta + o(1)}, \qquad S = X^{s + o(1)}, \qquad \frac{1}{2} \le \beta < 1. \tag{71}$$

On one denominator and exact-conductor cell let

$$Y = X^{y + o(1)}, \qquad F_i = X^{f_i + o(1)}, \qquad k_i = \max\{f_i, \beta - f_i\}. \tag{72}$$

The full rectangle gives

$$0 \le y \le 2s, \qquad 0 \le f_1, f_2 \le s. \tag{73}$$

Relative to the natural exponent $1 + \beta$, the norm and Gauss routes in (70) have savings

$$\eta_A(y, f_1, f_2) = \frac{\beta + 1 - y - k_1 - k_2}{2}, \tag{74}$$

$$\eta_B(f_1, f_2) = 1 - f_1 - f_2 - \frac{k_1 + k_2}{2}. \tag{75}$$

Since decreasing y improves η_A, define

$$M_\beta(s) := \inf_{0 \le f_1, f_2 \le s} \max\{\eta_A(2s, f_1, f_2), \eta_B(f_1, f_2)\}. \tag{76}$$

Proposition 4.1 (Exact relaxed-box base minimax)**.** *For $1/2 \le \beta < 1$,*

$$\boxed{M_\beta(s) = \begin{cases} 1 - \beta - s, & 0 \le s \le (1-\beta)/2, \\ \dfrac{3 - 3\beta - 2s}{4}, & (1-\beta)/2 \le s \le \beta/2, \\ \dfrac{3 - \beta - 6s}{4}, & \beta/2 \le s \le (3\beta - 1)/2, \\ \dfrac{\beta + 1 - 4s}{2}, & s \ge (3\beta - 1)/2. \end{cases}} \tag{77}$$

The formula is exact for the relaxed box (73); occupied arithmetic cells form a subset.

Proof. This is the same two-variable convex minimax that appears in the annular joint ledger, with the upper endpoint now equal to the base exponent s [6]. We include the optimization to make the initial-value theorem self-contained.

If $f_i \le \beta/2$, put $\sigma = f_1 + f_2$. Then

$$\eta_A = \frac{1 - \beta - 2s + \sigma}{2}, \qquad \eta_B = 1 - \beta - \frac{\sigma}{2}. \tag{78}$$

They meet at

$$\sigma_* = s + \frac{1-\beta}{2}, \qquad \eta_A = \eta_B = \frac{3 - 3\beta - 2s}{4}. \tag{79}$$

For $s \le (1-\beta)/2$, this point lies beyond $0 \le \sigma \le 2s$, and the endpoint gives $1 - \beta - s$. It is available for $(1-\beta)/2 \le s \le \beta/2$.

Now suppose $s \ge \beta/2$. On the high–high corner $f_1 = f_2 = s$, the two routes give

$$\eta_A = \frac{\beta + 1 - 4s}{2}, \qquad \eta_B = 1 - 3s. \tag{80}$$

Here

$$\eta_A - \eta_B = \frac{\beta - 1 + 2s}{2} \ge 0,$$

so the relevant upper envelope is the first expression. In a mixed cell write $x = f_{\text{low}} \le \beta/2$ and $z = f_{\text{high}} \ge \beta/2$. Both savings decrease with z, so $z = s$. The two remaining affine functions meet at

$$x_* = \frac{1-\beta}{2}, \qquad \eta_A = \eta_B = \frac{3 - \beta - 6s}{4}. \tag{81}$$

While the low–low intersection remains available, its value exceeds the mixed value by

$$\frac{3 - 3\beta - 2s}{4} - \frac{3 - \beta - 6s}{4} = s - \frac{\beta}{2} \ge 0.$$

The high–high and mixed faces cross at $s = (3\beta - 1)/2$. Beyond that point the low–low endpoint is $1 - 3\beta/2$, which is also no smaller than the high–high face. This proves the last two branches. Continuity at every boundary is immediate. ☐

Remark 4.2 (Interpretation of the minimax). A nonpositive value means that the two displayed cell estimates do not give a uniform power saving over the relaxed box. It is not an arithmetic impossibility theorem. Conversely, a positive value is a valid lower bound for every occupied cell, whether or not all equality witnesses occur arithmetically.

4.2 Principal mean and Fourier tail

For fixed q, assemble the raw full-base unit fiber before taking any conductor projection:

$$Z_{q;S}^{\mathrm{base}}(\xi) := \sum_{\substack{u,v\le S\\ \mathrm{lcm}(u,v)=q}} b_R(u)b_R(v)Z_{u,v}(\xi). \tag{82}$$

For one Mellin parameter, write $G=\Phi_{\boldsymbol{\tau}}$ and define the nonzero principal-mean contribution at denominator q by

$$\begin{aligned}\mathscr{I}_{q;S,\boldsymbol{\tau}}^{\mathrm{mean},\neq 0} := \frac{J}{Hq}\sum_{r\neq 0} c_H(r)c_q(r)\widehat{G}\left(\frac{rJ}{Hq}\right)\\ \times \sum_{\substack{u,v\le S\\ \mathrm{lcm}(u,v)=q}} b_R(u)b_R(v)\overline{Z}_{u,v}.\end{aligned} \tag{83}$$

Lemma 4.3 (Global principal mean)**.** *For some fixed $C_{\mathscr{D}}>0$,*

$$\sum_q |\mathscr{I}_{q;S,\boldsymbol{\tau}}^{\mathrm{mean},\neq 0}| \ll_{\varepsilon,\mathscr{D}} X^{\varepsilon}(1+|\boldsymbol{\tau}|)^{C_{\mathscr{D}}}\|x_{\boldsymbol{\tau}}\|_1\|y_{\boldsymbol{\tau}}\|_1, \tag{84}$$

$$\int\sum_q |\mathscr{I}_{q;S,\boldsymbol{\tau}}^{\mathrm{mean},\neq 0}|\, d|\nu|(\boldsymbol{\tau}) \ll_{\varepsilon,\mathscr{D}} Q^2X^{\varepsilon}. \tag{85}$$

This includes $u=v=1$; cells with exactly one principal conductor coordinate remain in theorem 3.5.

Proof. The unit-fiber mean satisfies

$$|\overline{Z}_{u,v}| \le \frac{\|x_{\boldsymbol{\tau}}\|_1\|y_{\boldsymbol{\tau}}\|_1}{\varphi(q)}, \qquad q=[u,v].$$

Put $a_q = J/(Hq)$. The inequality $|c_q(r)|\le \sum_{d|(q,r)} d$, Schwartz decay, and the divisor bound give

$$a_q\sum_{r\neq 0}|c_H(r)c_q(r)||\widehat{G}(a_qr)| \ll_{\varepsilon,\mathscr{D}} (1+|\boldsymbol{\tau}|)^{C_{\mathscr{D}}}X^{\varepsilon}.$$

Finally,

$$\sum_{u,v\le S}\frac{1}{[u,v]} \ll (\log(2S))^3, \tag{86}$$

while $1/\varphi(q)\ll_{\varepsilon} X^{\varepsilon}/q$ and (23) absorbs the divisor coefficients. This proves (84). Integrate with (43) to obtain (85). When $q=1$, the nonzero frequencies are even rapidly decreasing because $a_1\asymp J$. □

The discrepancy is projected inside each genuine unit fiber and then summed over the $O(\log^2(2S))$ conductor cells. The retained part is bounded by theorem 3.5. The unprojected high-frequency tail is removed before this pairwise triangle inequality.

Lemma 4.4 (Automatic physical Fourier tail)**.** *For every prescribed $A>0$, the complete raw-base frequency range*

$$|r| > X^{\omega}Hq/J \tag{87}$$

contributes $O_{A,\mathscr{D}}(X^{-A}Q^2)$ after fixed-residue and physical Mellin reassembly.

Proof. For one component and divisor pair, delete the row mask in absolute value and use the ℓ^1 fiber mass $\sum_\xi |Z_{u,v}(\xi)| \le \|x\|_1\|y\|_1$. Repeated integration by parts gives, for every N,

$$|\widehat{\Phi}_{\boldsymbol{\tau}}(y)| \ll_N (1+|\boldsymbol{\tau}|)^N(1+|y|)^{-N}.$$

With $a_q = J/(Hq)$, an integral comparison yields uniformly in q

$$a_q \sum_{|r|>X^\omega/a_q} (1+a_q|r|)^{-N} \ll_N X^{-\omega(N-1)}. \tag{88}$$

The prefactor therefore cancels the frequency-lattice spacing. By (23), the total divisor coefficient mass on $u, v \le S$ is at most a fixed power of X. Choose N after A, integrate with (43), and sum the fixed residue cells. □

4.3 The raw nonzero base theorem

Let $(\mathcal{Q}_S^{\mathsf{AA}})^{\neq 0}$ denote the physical sum of all nonzero additive frequencies from $\mathsf{A}_{m_\alpha,S}\mathsf{A}_{m_\gamma,S}$, with the same row family, mask, periodic factor, and smooth weight as in (31).

Theorem 4.5 (Physical raw-joint nonzero closure)**.** *Assume the physical interface of section 2, the continuous transfer (41)–(43), the no-wrap condition (60), the fixed-power cutoff $S \le X^{C_S}$ from (14), and*

$$M_\beta(s) > 0 \tag{89}$$

with a fixed margin. Then some fixed $\eta > 0$ satisfies

$$\boxed{(\mathcal{Q}_S^{\mathsf{AA}})^{\neq 0} \ll_{\mathscr{D}} XQX^{-\eta}.} \tag{90}$$

No condition $S < J$ is used.

Proof. For each dyadic Y, F_1, F_2, apply the physically integrated estimate (70). The exponent calculation (74)–(76) and proposition 4.1 give a common fixed-power saving over all nonprincipal occupied cells. The logarithmically many cells are absorbed by choosing $\eta < M_\beta(s)$ after the X^ε losses.

The principal mean (85) saves the fixed power $J = X^{1-\beta+o(1)}$ against $XQ = JQ^2$. Remove the unprojected Fourier tail with lemma 4.4, restore the fixed residue cells, and note that the component integration has already been performed in (70), (85), and the proof of lemma 4.4. Thus one may take any sufficiently small fixed η below both $M_\beta(s)$ and $1-\beta$, after the auxiliary losses are fixed. □

Remark 4.6 (The removed old face)*.* The face $1 - 3\beta/2$ in (5) belongs to the earlier centered base decomposition. It does not appear in the raw joint minimax (77). This proves that the face is not an intrinsic boundary of the full-rectangle joint conductor method; it does not remove any barrier outside this selected packet.

5 Drift restoration and the Poisson-long base theorem

The raw joint nonzero theorem estimates only the coefficient-one term in (7). We now restore the two drift legs and the constant term. The key point is that $\delta_H(m)$ has a fixed-power saving, so a soft target-divisor bound is sufficient.

5.1 A uniform target-divisor bound

Lemma 5.1 (Pointwise raw-prefix bound). *For every fixed $\varepsilon > 0$, uniformly on the physical orbit support,*

$$\boxed{|\mathsf{A}_{m,S}(j)| \ll_{\varepsilon,\mathscr{D}} X^{\varepsilon}.} \tag{91}$$

The estimate is independent of the comparison between S and J.

Proof. Put $N = mj + h$. By (17), $1 \le |N| \ll X^{C_0}$ for a fixed C_0. Every effective u in (25) divides N. Hence

$$|\mathsf{A}_{m,S}(j)| \le \tau(|N|) \max_{\substack{u|N \\ u \le S}} |b_R(u)|. \tag{92}$$

Use (23) and the classical divisor bound $\tau(|N|) \ll_{\varepsilon,C_0} X^{\varepsilon}$, after reducing the auxiliary exponent. The cutoff $u \le S$ only deletes divisors. □

Remark 5.2 (Why the target cannot be zero). If $N = 0$, every positive integer would divide N, and lemma 5.1 would be false. The physical target support (17) explicitly excludes this case. Equivalently, in the underlying positive dyadic packet one has $mj + h \asymp X$.

Let $\mathcal{D}_S^{\mathrm{phys}}$ denote the physical contribution of the last three terms in

$$P_{m,S}P_{n,S} = \mathsf{A}_{m,S}\mathsf{A}_{n,S} - \delta_H(m)\mathsf{A}_{n,S} - \delta_H(n)\mathsf{A}_{m,S} + \delta_H(m)\delta_H(n). \tag{93}$$

It is the complete sampled orbit contribution, before any additive frequency projection.

Proposition 5.3 (Full and zero-frequency physical drift bounds). *For every fixed $\varepsilon > 0$,*

$$\boxed{|\mathcal{D}_S^{\mathrm{phys}}| + |(\mathcal{D}_S^{\mathrm{phys}})^0| \ll_{\varepsilon,\mathscr{D}} JQ^2 \left(L^{-1}X^{\varepsilon} + L^{-2}X^{\varepsilon}\right).} \tag{94}$$

Consequently, under $L \ge X^{\lambda_0}$, the drift has a fixed-power saving relative to $XQ \asymp JQ^2$.

Proof. There are $O_{\mathscr{D}}(J)$ orbit points. Delete the bounded mask, periodic factor, and smooth weight. Apply lemma 5.1, the drift estimate (24), and the two physical row ℓ^1-bounds (20). Each mixed drift leg is at most

$$J \cdot Q \cdot Q \cdot L^{-1}X^{\varepsilon}.$$

The constant term is at most $JQ^2L^{-2}X^{\varepsilon}$.

For the additive zero term, the complete-period mean of $\mathsf{A}_{m,S}$ is $D_{m,S}$. Directly from (26), (23), and $S \le X^{C_S}$,

$$|D_{m,S}| \ll_{\mathscr{D}} (\log(2X))^{O(1)} \ll_{\varepsilon} X^{\varepsilon}.$$

The bounded zero Fourier functional, the same row ℓ^1 estimates, and (24) therefore give the identical bound for $|(\mathcal{D}_S^{\mathrm{phys}})^0|$. Renaming the auxiliary exponent proves (94). □

Remark 5.4 (Comparison with the centered scalar estimate). The soft factor X^{ε} in (91) cannot be multiplied by $E_S(m) \ll_A (\log X)^{-A}$ and then absorbed by choosing A. This is why the proof must use $P = Z + E$ only at frequency zero and $P = \mathsf{A} - \delta_H$ at nonzero frequency. The fixed-power factor L^{-1} in δ_H does absorb the soft divisor loss.

5.2 Frequency-dependent assembly

For each finite divisor pair, fixed residue cell, and smooth Mellin component, Poisson summation decomposes the sampled orbit into its additive zero term and nonzero terms. By linearity,

$$\mathcal{Q}_S^{\mathrm{phys}} = (\mathcal{Q}_S^{\mathrm{phys}})^0 + (\mathcal{Q}_S^{\mathsf{AA}})^{\neq 0} + (\mathcal{D}_S^{\mathrm{phys}})^{\neq 0}. \tag{95}$$

The first term is the zero frequency of the complete calibrated product and is evaluated with $P = Z + E$. The last two terms use $P = \mathsf{A} - \delta_H$. In particular,

$$|(\mathcal{D}_S^{\mathrm{phys}})^{\neq 0}| \leq |\mathcal{D}_S^{\mathrm{phys}}| + |(\mathcal{D}_S^{\mathrm{phys}})^0|, \tag{96}$$

where the right side is bounded in proposition 5.3. This formula does not double-count the zero frequency: the first term already contains the exact recombination (36).

Theorem 5.5 (Calibrated Poisson-long base closure)**.** *Retain the physical opened-row provenance, row norms, common smooth weight, divisor-independent off-diagonal mask, and nonzero target support of section 2. Assume $R \leq S \leq X^{C_S}$, $L \geq X^{\lambda_0}$, the calibration and Ramanujan-square inputs (29), (39), the continuous transfer (41)–(43), and the no-wrap margin (60). Suppose*

$$M_\beta(s) > 0, \qquad Q = X^{\beta+o(1)}, \quad S = X^{s+o(1)}, \quad \frac{1}{2} \leq \beta < 1, \tag{97}$$

with fixed margins. Then for every fixed $B > 0$,

$$\boxed{\mathcal{Q}_S^{\mathrm{phys}} \ll_{B,\mathscr{D}} XQ(\log X)^{-B}.} \tag{98}$$

The theorem permits $S > J$; no Poisson-short hypothesis is present.

Proof. Use the exact assembly (95). The complete calibrated zero term has arbitrary logarithmic saving by theorem 2.3. The raw joint nonzero term has a fixed power saving by theorem 4.5. The remaining nonzero drift is bounded by proposition 5.3. Choose the auxiliary $\varepsilon < \lambda_0/3$; the drift then also has a fixed-power saving. Each power-saving contribution is $O_B\Big(XQ(\log X)^{-B}\Big)$ for every fixed B and sufficiently large X. The fixed dyadic, sign, content, and residue reassembly costs are absorbed by choosing the zero calibration order after B. □

Corollary 5.6 (Beyond the old Poisson boundary)**.** *If the hypotheses of theorem 5.5 hold together with*

$$s > 1 - \beta, \tag{99}$$

then (98) is a calibrated initial-value estimate whose divisor cutoff is strictly longer than the orbit length $J = X^{1-\beta+o(1)}$.

Remark 5.7 (What has and has not been bypassed). The theorem bypasses the Poisson-short condition only for the selected raw joint base packet. It does so through exact zero recombination and row/conductor averaging. It does not prove a uniform two-point Möbius theorem for the reflected ultra-long complement.

6 The high-β initial value and upper splice

We now verify the exact source-compatible sample that was left with an open initial value in TPC-26 and TPC-27.

6.1 Source geometry

Put

$$\sigma = \frac{1}{10000}, \qquad \lambda = \frac{10}{21} - \sigma = \frac{99979}{210000}, \qquad \delta = \frac{7}{60}. \tag{100}$$

Take

$$\beta = \frac{267}{400}, \qquad s = \frac{23}{60}, \qquad t = \frac{193}{500}. \tag{101}$$

Then

$$R = S = X^{23/60+o(1)}, \qquad J = X^{133/400+o(1)}, \qquad T = X^{193/500+o(1)}. \tag{102}$$

The opened-d exponent and its available endpoint are

$$d_* = \beta - \lambda = \frac{10049}{52500}, \qquad v_* = \frac{1}{4} - \frac{\delta}{2} = \frac{23}{120}, \tag{103}$$

with

$$v_* - d_* = \frac{9}{35000} > 0. \tag{104}$$

In the notation of TPC-26, the source relations $J = K/D$, $D_0 < D \le V$, and $K > 2V$ were verified for these exact exponents [6]. We also have

$$\lambda - t = \frac{18919}{210000} > 0, \tag{105}$$

so $L > 2T \ge 2S$ for all sufficiently large X. The present proof requires only the weaker fixed-power condition $L \ge X^{\lambda_0}$, but (105) confirms the original source geometry.

The base lies strictly beyond the Poisson length:

$$s - (1-\beta) = \frac{23}{60} - \frac{133}{400} = \frac{61}{1200} > 0. \tag{106}$$

Thus the old Poisson-short base theorem cannot be invoked.

6.2 Old and new base ledgers

The three entries in the earlier centered base ledger are

$$\left(1 - \frac{3\beta}{2}, \frac{\beta+1-4s}{2}, \frac{3-\beta-6s}{4}\right) = \left(-\frac{1}{800}, \frac{161}{2400}, \frac{13}{1600}\right). \tag{107}$$

The first face is negative. This records the exact failure of the old centered nonzero theorem; it is not a failure of the zero theorem.

For the full raw rectangle,

$$\frac{\beta}{2} = \frac{267}{800} < s = \frac{23}{60} < \frac{3\beta-1}{2} = \frac{401}{800}. \tag{108}$$

Therefore the third branch of (77) applies and gives

$$\boxed{M_{267/400}(23/60) = \frac{3 - 267/400 - 6(23/60)}{4} = \frac{13}{1600}.} \tag{109}$$

The competing high–high face is $161/2400$, so the mixed witness is the actual relaxed-box bottleneck.

The drift has the independent fixed-power factor L^{-1}. At this sample the source exponent is

$$\lambda = \frac{99979}{210000} > 0, \tag{110}$$

and proposition 5.3 applies after choosing its soft X^{ε} loss below this fixed value. In particular, no compensation for the Poisson overrun $61/1200$ is required.

Corollary 6.1 (Closed high-β initial value)**.** *For the actual opened-row packet with the fixed primitive, mask, no-wrap, smooth-transfer, and target-support interfaces above, the parameters* (100)–(101) *satisfy*

$$\boxed{\mathcal{Q}_S^{\mathrm{phys}} \ll_{B,\mathscr{D}} XQ(\log X)^{-B}} \tag{111}$$

for every fixed $B > 0$, although $S > J$.

Proof. The exact source relations give the physical row interface. Equation (109) supplies the positive full-rectangle conductor margin, while (110) supplies the existing fixed-power row-prime margin. Apply theorem 5.5. □

6.3 Splicing to the upper cutoff

TPC-26 proves annular cutoff stability between these same S and T whenever $M_\beta(t) > 0$. Here

$$\frac{\beta}{2} < t < \frac{3\beta - 1}{2}, \tag{112}$$

and the two upper faces are

$$\frac{\beta + 1 - 4t}{2} = \frac{247}{4000}, \qquad \frac{3 - \beta - 6t}{4} = \frac{33}{8000}. \tag{113}$$

Thus

$$\boxed{M_{267/400}(193/500) = \frac{33}{8000}.} \tag{114}$$

Also $t - s = 1/375 > 0$. The imported theorem gives

$$\mathcal{Q}_T^{\mathrm{phys}} - \mathcal{Q}_S^{\mathrm{phys}} \ll_{B,\mathscr{D}} XQ(\log X)^{-B}. \tag{115}$$

Its remaining hypotheses use the same physical packet, common weight, mask, no-wrap, and source interfaces retained above. Moreover $1 - \beta = 133/400 > 0$, $S < T$, and (105) provide the required fixed-power support and separation margins.

Theorem 6.2 (Closed high-β selected truncated square)**.** *Under the physical interfaces above, the exact sample* (100)–(101) *satisfies, for every fixed $B > 0$,*

$$\boxed{\mathcal{Q}_T^{\mathrm{phys}} \ll_{B,\mathscr{D}} XQ(\log X)^{-B}.} \tag{116}$$

Proof. Use the exact identity

$$\mathcal{Q}_T^{\mathrm{phys}} = \mathcal{Q}_S^{\mathrm{phys}} + (\mathcal{Q}_T^{\mathrm{phys}} - \mathcal{Q}_S^{\mathrm{phys}})$$

and combine corollary 6.1 with (115). □

6.4 Exact scope of the splice

The conclusion estimates one selected calibrated square truncated at T. It does not estimate the ultra-long complement

$$\sum_{\substack{T < u \le U \\ u \mid mj+h}} a(u), \tag{117}$$

nor the products involving that complement. Writing the reflected quotient variables as $mj + h = ur'$ and $nj + h = vs'$, divisor reflection there produces the exact determinant surface

$$n\,u\,r' - m\,v\,s' = h(n - m), \tag{118}$$

together with orbit congruences and growing two-point Möbius coefficients. No estimate for that surface is asserted here. The separate positivity/main-term stage is also untouched.

Table 1: Exact margins in the high-β selected-square splice.

Gate	Exact value	Status
Poisson overrun $s - (1 - \beta)$	61/1200	base is long
Old centered face $1 - 3\beta/2$	$-1/800$	old route fails
Full raw-base minimax $M_\beta(s)$	13/1600	closed here
Row-prime exponent λ	99979/210000	drift is power-small
Annular minimax $M_\beta(t)$	33/8000	imported from TPC-26

7 Exact certificate, dependency ledger, and scope

The proof has one new finite algebraic identity, one new analytic full-rectangle theorem, and one exact rational sample. We separate these from imported interfaces and from statements that remain open.

7.1 Exact finite certificate

The standard-library script

`experiments/tpc28_certificate.py`

uses rational arithmetic and formal prime-log polynomials. It checks:

(i) the pointwise raw/drift product identity for finite opened rows;

(ii) the direct connected double sum and complete-period raw, centered, and calibrated zero kernels, including (36) and a case with $\operatorname{rad}|h| \neq |h|$;

(iii) CRT compatibility and primitive complete-period densities for full-base divisor pairs, including pairs with one unit divisor;

(iv) the two-route minimax lower bound on exact rational grids and the equality witnesses on every piece of (77);

(v) all rational source, Poisson-overrun, base, drift, and upper margins in table 1.

Every assertion uses an exception-based checker, so running Python with optimization cannot delete a test. The script writes a deterministic JSON summary and hashes its normalized source and a canonical payload. It checks finite algebra and rational bookkeeping only; it is not numerical evidence for an asymptotic Möbius or prime-pair claim. Both ordinary and optimized Python complete 33,299 exact checks and regenerate byte-identical JSON. The final hashes are

- source SHA-256: `c3ad5c993e1823a4752c5c31ef5e348342982607c078e8bc4343697d5dad9948`;
- JSON SHA-256: `e75a92e2eb78ce297fa305e5c8ea42eab0a6cec591e3ff989eb5566bd90ab6ec`;
- certificate digest: `3cb8458021bd05d8b04154137a92a068633944679a038db9c5e4f342dff563a3`.

7.2 Dependency ledger

Table 2: Logical status of the ingredients used in TPC-28.

Ingredient	Status	Role
Opened-row ℓ^1/ℓ^2 provenance	Imported	Gives physical row mass and continuous Mellin transfer; TPC-25.
Uniform prefix calibration $D = \delta + E$	Imported	Gives arbitrary-logarithmic $E_S(m)$; TPC-20 and TPC-27.
Ramanujan-square bound for K_S^0	Imported	Closes the connected physical zero mode for polynomial S; TPC-27.
Raw/centered calibrated zero identity	New here	Cancels the two $E_S Z_S$ zero legs exactly.
Full-base tensor energy and cell bound	New here	Extends the joint conductor mechanism to $u, v \le S$ without an annular lower cutoff.
Pointwise target-divisor drift bound	New here	Restores nonzero scalar legs with the fixed-power L^{-1} factor.
Upper annular cutoff stability	Imported	Splices the newly closed base S to T; TPC-26.
High-β initial and upper values	New here	Closes the selected source-compatible truncated square.

The full-base conductor proof uses the classical exact-conductor large sieve, not an unproved distribution conjecture. The zero theorem uses the proved uniform Möbius calibration and the exact finite Ramanujan-square factorization. Thus theorems 5.5 and 6.2 require no unproved conjecture beyond their explicitly stated physical packet hypotheses.

7.3 Three invalid shortcuts

The proof also identifies three tempting but invalid replacements:

(1) One cannot estimate nonzero EZ by multiplying $E_S \ll_A (\log X)^{-A}$ with the soft bound $Z \ll X^{\varepsilon}$. A fixed X^{ε} is not absorbed by choosing A. The nonzero decomposition must use the power-small drift δ_H.

(2) One cannot obtain the full-base theorem by formally setting the lower endpoint of an annular prefix theorem below R. The full unit fibers, energy, principal mean, and reassembly must be defined directly, as in sections 3 and 4.

(3) One cannot conductor-project the coherent additive zero terms before their raw/centered recombination. In particular $b_R(1)$ is generally nonzero. The zero identity (36) must be applied first.

7.4 What is new and what remains open

The new conclusions are:

- a full-rectangle nonzero conductor theorem governed by $M_\beta(s)$;
- calibrated base closure without $S < J$;
- closure of the particular source-compatible $\beta = 267/400 > 2/3$ initial value;
- after importing TPC-26 cutoff stability, closure of the corresponding selected square through $T = X^{193/500+o(1)}$.

The following are not conclusions of this paper:

- the ultra-long complement beyond T;
- the full TPC-18 residual or a complete original prime-pair correlation;
- a polynomially uniform affine two-point Möbius theorem on the reflected surface (118);
- positivity, a Hardy–Littlewood asymptotic, or a twin-prime lower bound;
- any general breach of the sieve parity barrier.

The next analytic gate is therefore cleaner than before: the selected high-β square no longer lacks an initial value. What remains on this branch is the reflected ultra-long complement, where the exact surface (118) must be combined with orbit congruences, sparse incidence geometry, and genuine cancellation in growing affine Möbius coefficients. A positive theorem, a precise conditional reduction, or a sharp barrier theorem at that gate would all be logically downstream of the present result.

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